



GROUP ASSIGNMENT
TECHNOLOGY PARK MALAYSIA
AICT009-4-2-IDA
INTRODUCTION TO DATA ANALYTICS

HAND OUT DATE: 22 MAY 2025

HAND IN DATE: 31 JULY 2025

WEIGHTAGE: 50%

INSTRUCTIONS TO CANDIDATES:

1. Submit your assignment at the administrative counter.
2. Students are advised to underpin their answers with the use of references (cited using the Harvard Name System of Referencing).
3. Late submissions will be awarded zero (0) unless Extenuating Circumstances (EC) are upheld.
4. Cases of plagiarism will be penalized.
5. The assignment should be bound in an appropriate style (comb bound or stapled).
6. Where the assignment should be submitted in both hardcopy and softcopy, the softcopy of the written assignment and source codes (where appropriate) should be on a CD in an envelope / CD cover and attached to the hardcopy.
7. You must obtain 50% overall to pass this module.

Title: Predictive Analytics for Diabetes Risk Using Health Indicators

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Intake Code: UCDF2407ICT(DI)

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Group Component

2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the characteristics, distribution, and the relationship between features collected within the Diabetes Health Indicators Dataset (Diabetes Health Indicator Dataset, 2022). This process helped identify key trends, validate assumptions, and guide feature selection for model development.

2.1. Initial Data Overview

The dataset contains **253,680 observations** with **22 numeric features**. The first few rows were examined, and the format and variable types were confirmed using `df.head()`. A further, more detailed look at the dataset was done with `df.info()`, revealing the information below:

- All columns are in the float64 format
- No missing values are identified (`df.isna().sum() = 0`)
- The target variable, `Diabetes_binary`, was originally in the form of four unique categories, by grouping the pre-diabetes and diabetes categories under the label 1.0 (the medical proceeding here was assumed appropriate)

Data Summary statistics from the `df.describe().T`:

- **BMI** was in the range of **12 and 98**, with a mean (28.38)
- **Age** was likewise measured in **13 categories**, with age distribution across older categories
- Number of mental health and physical health days asked was also desirable (as it was likely quality-of-life stats)

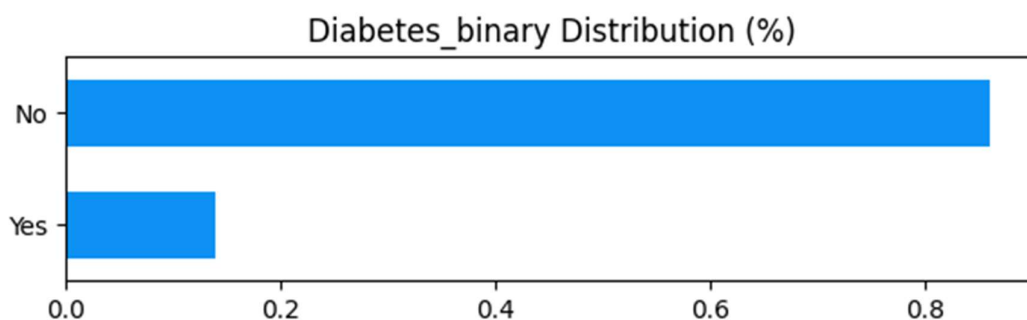


Figure 1 Distribution of Diabetes_binary as a Bar Chart

2.2. Binary Feature Exploration

For binary features like **HighBP**, **Smoker**, **HeartDiseaseorAttack**, and **PhysActivity**, the value counts and pie charts are segmented by diabetes status.

Notable findings were as follows:

- Hypertension (**HighBP**) and smoking status were higher in diabetics
- Physical activity (**PhysActivity**) was moderately lower in diabetics
- Pretty much all diabetic respondents had access to health care, but the cost barrier (**NoDochcCost**) was a minority of them.

These insights were useful in the next steps in directing the feature prioritization.

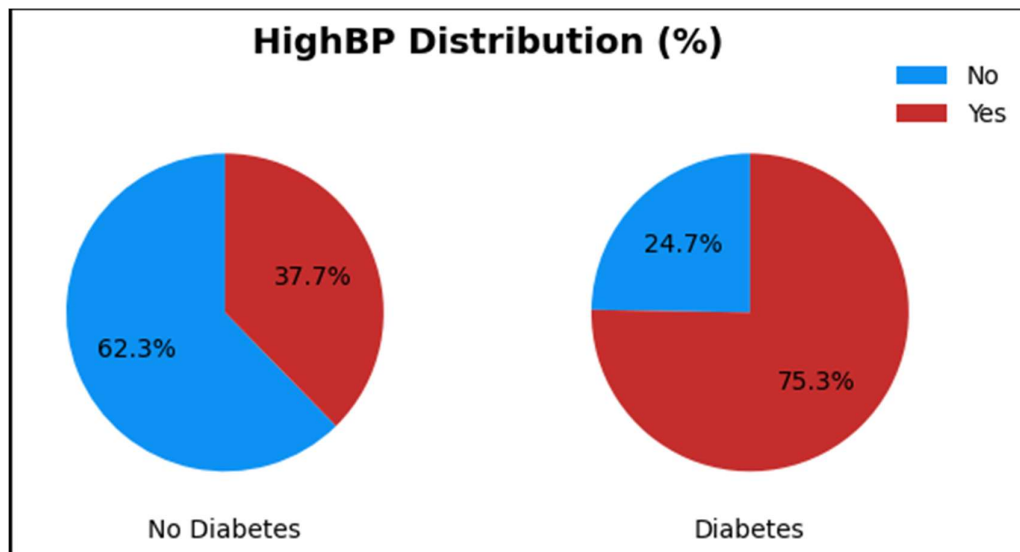


Figure 2 Pie Chart for HighBP Distribution

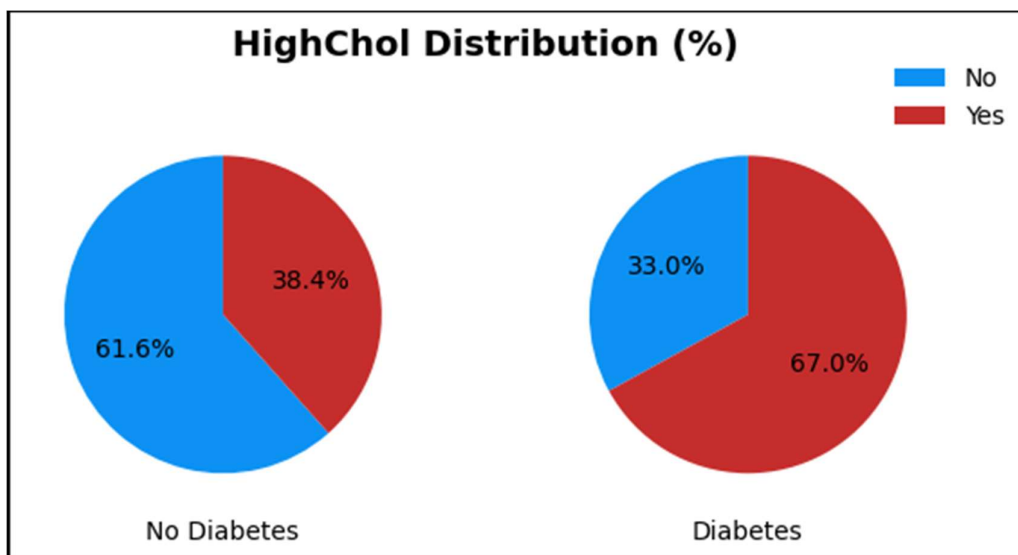


Figure 3 Pie Chart for HighChol Distribution

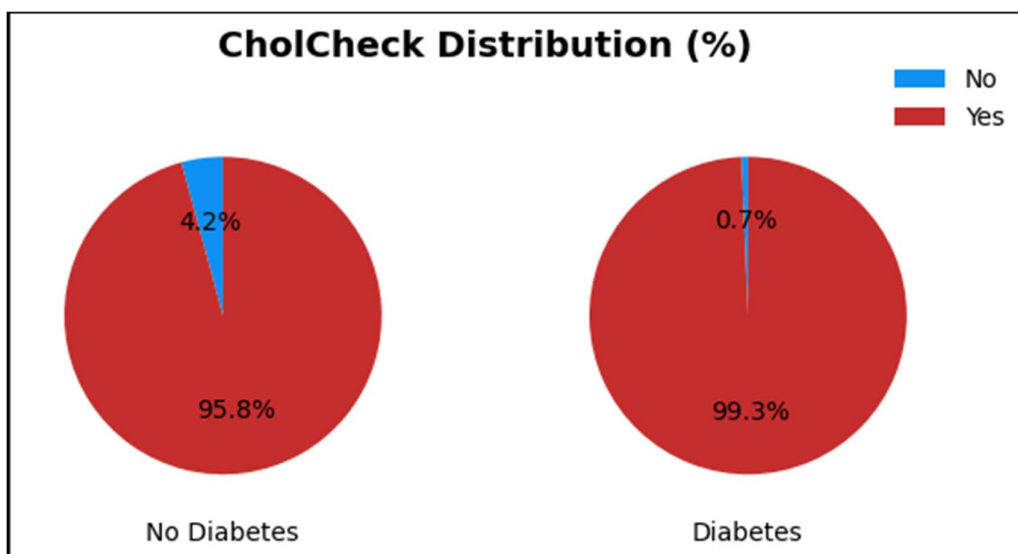


Figure 4 Pie Chart for CholCheck Distribution

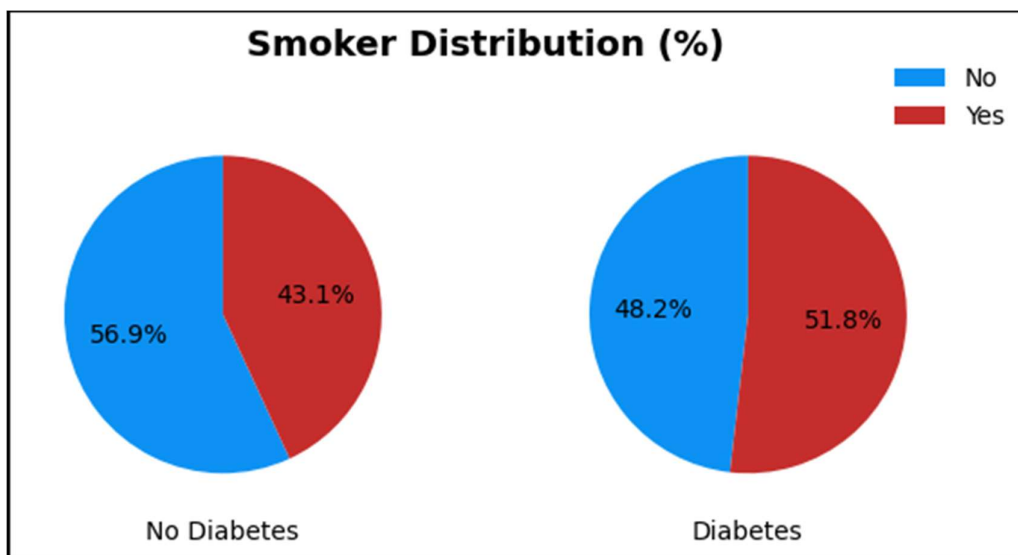


Figure 5 Pie Chart for Smoker Distribution

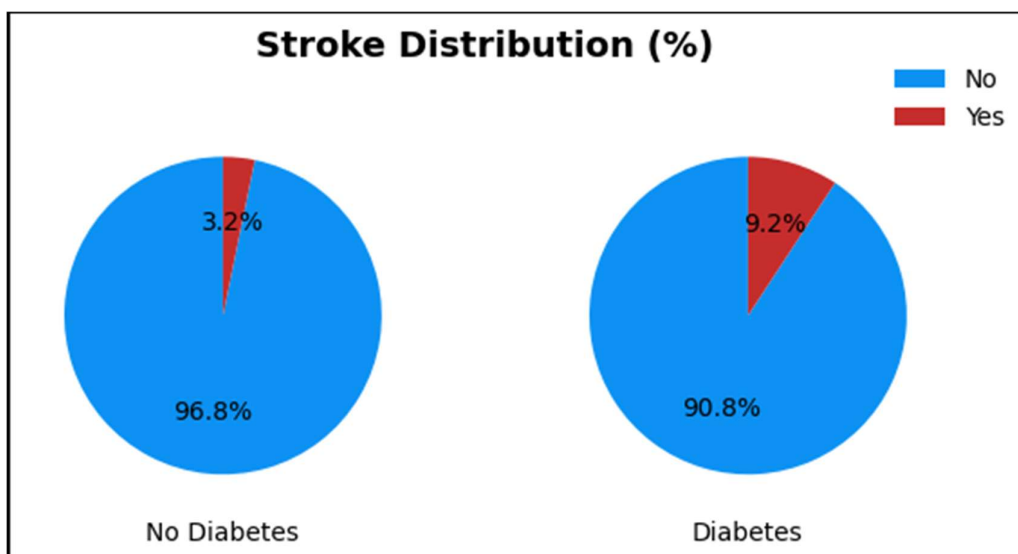


Figure 6 Pie Chart for Stroke Distribution

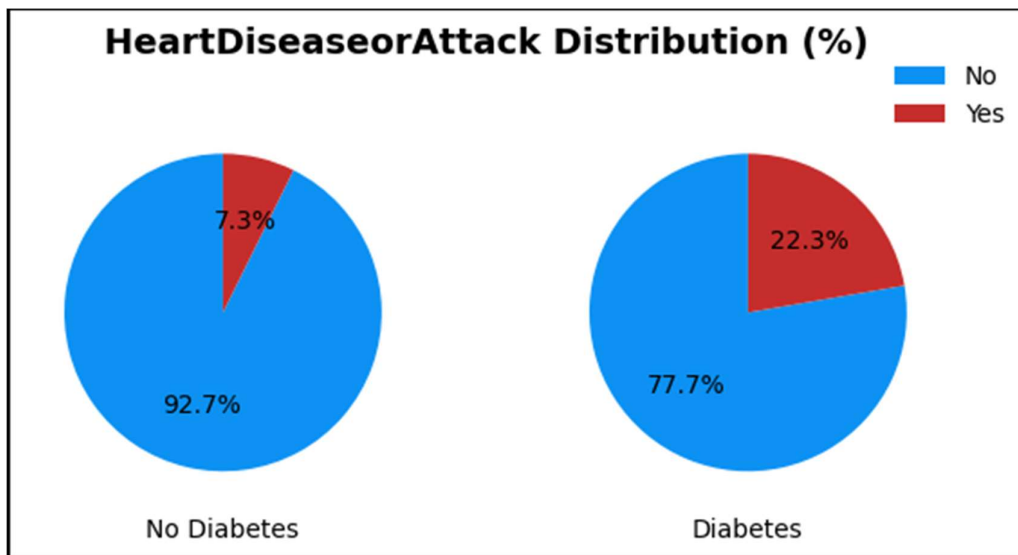


Figure 7 Pie Chart for HeartDiseaseorAttack Distribution

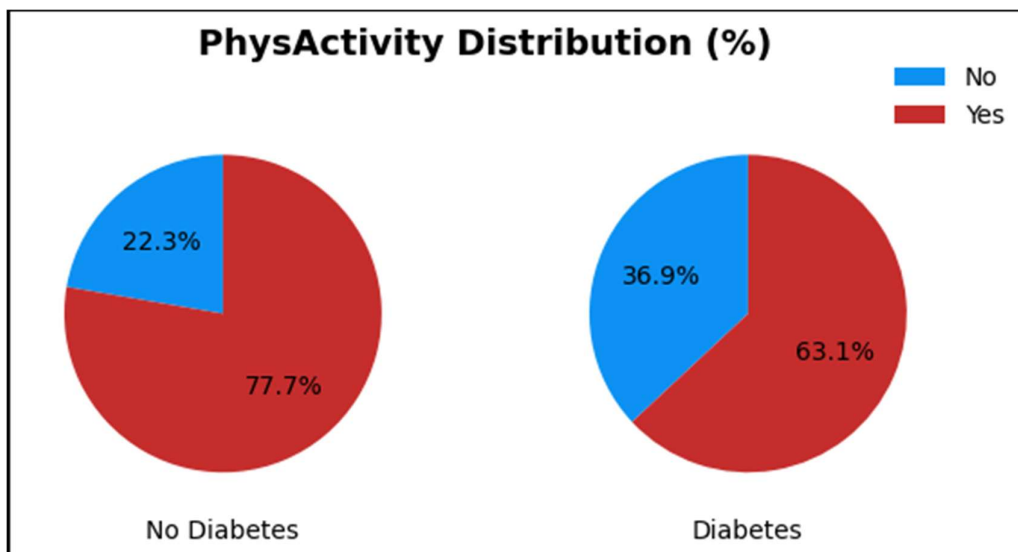


Figure 8 Pie Chart for PhysActivity Distribution

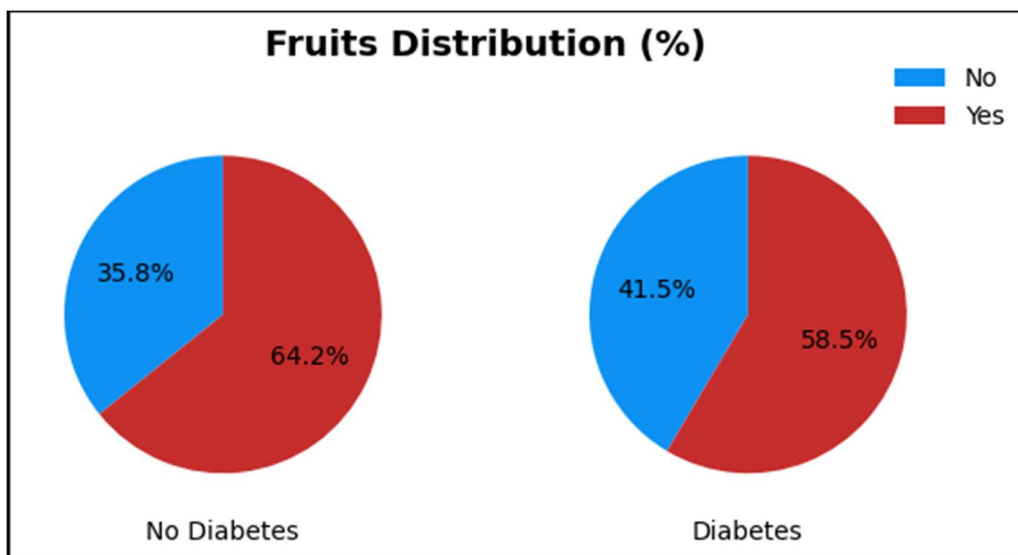


Figure 9 Pie Chart for Fruits Distribution

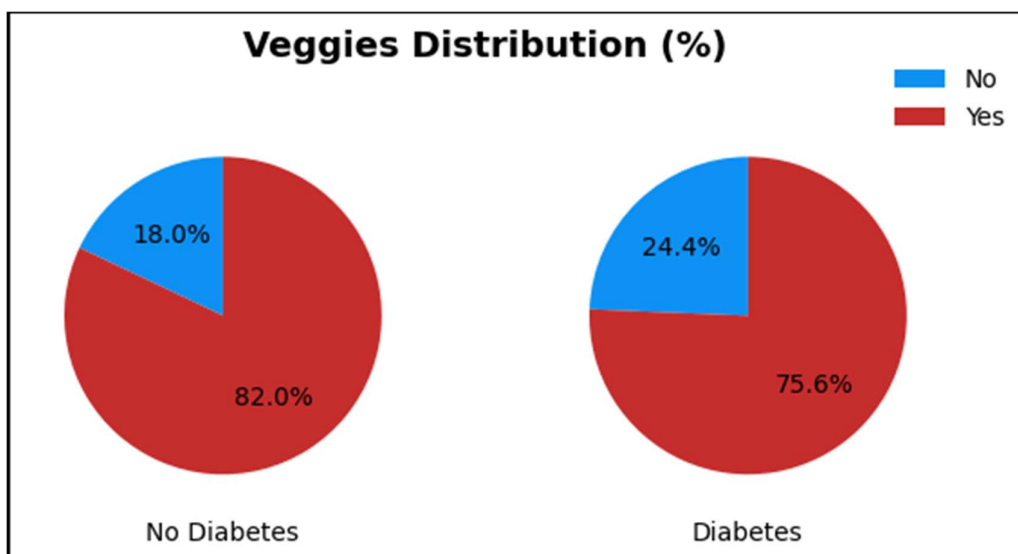


Figure 10 Pie Chart for Veggies Distribution

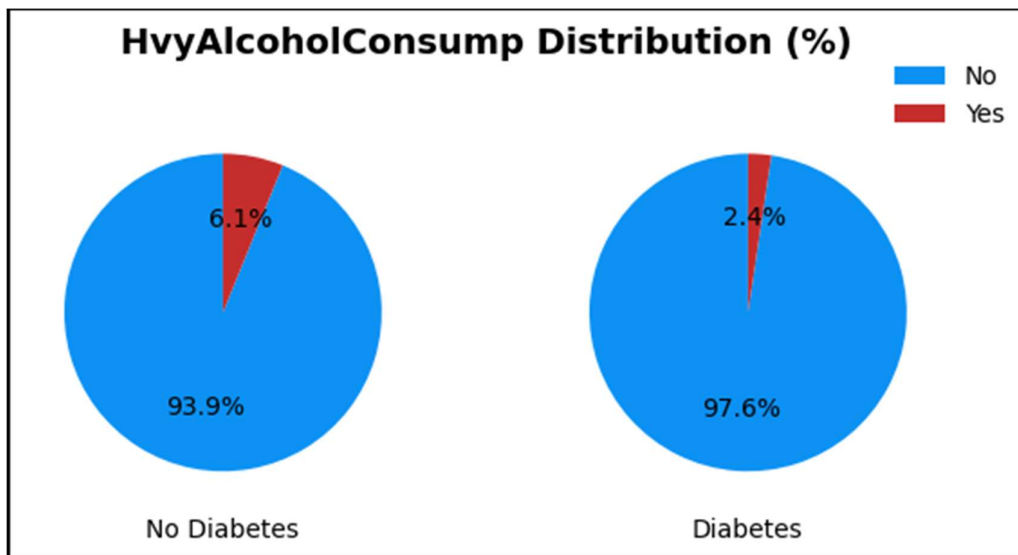


Figure 11 Pie Chart for HvyAlcoholConsump Distribution

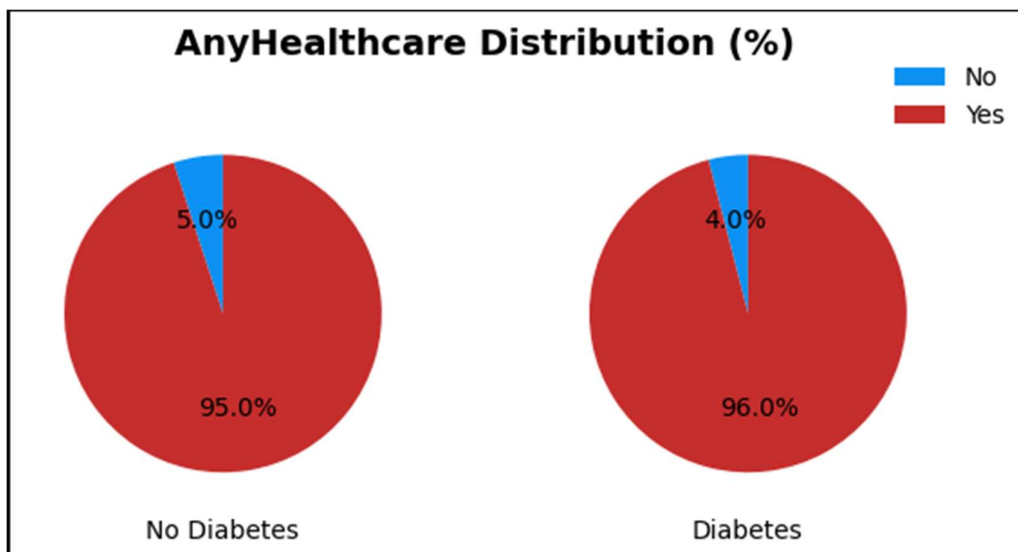


Figure 12 Pie Chart for AnyHealthcare Distribution

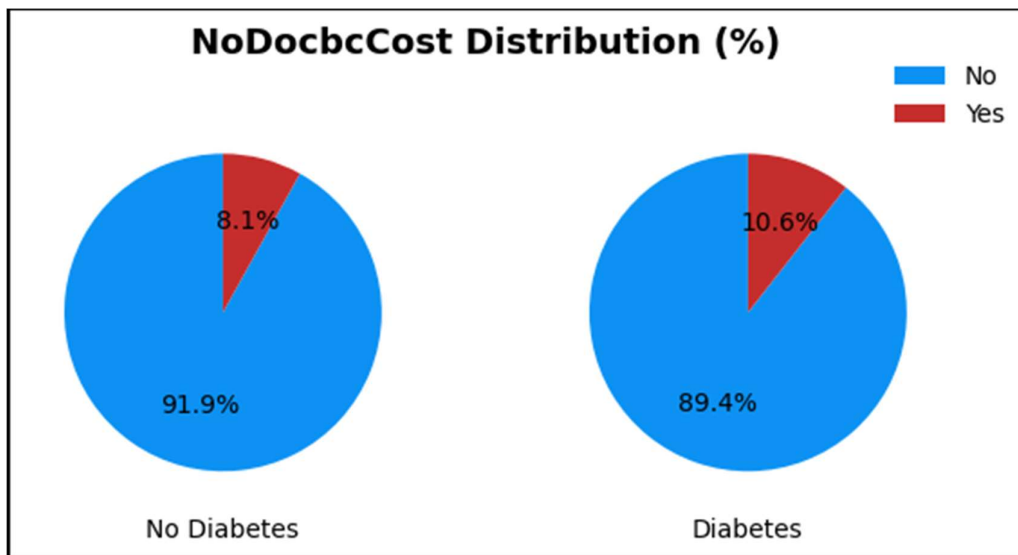


Figure 13 Pie Chart for NoDocbcCost Distribution

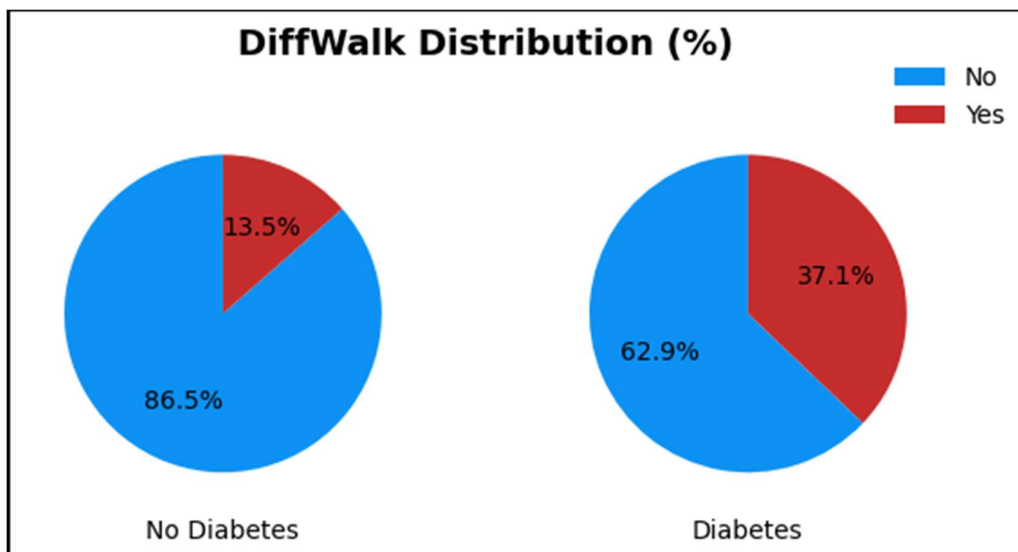


Figure 14 Pie Chart for DiffWalk Distribution

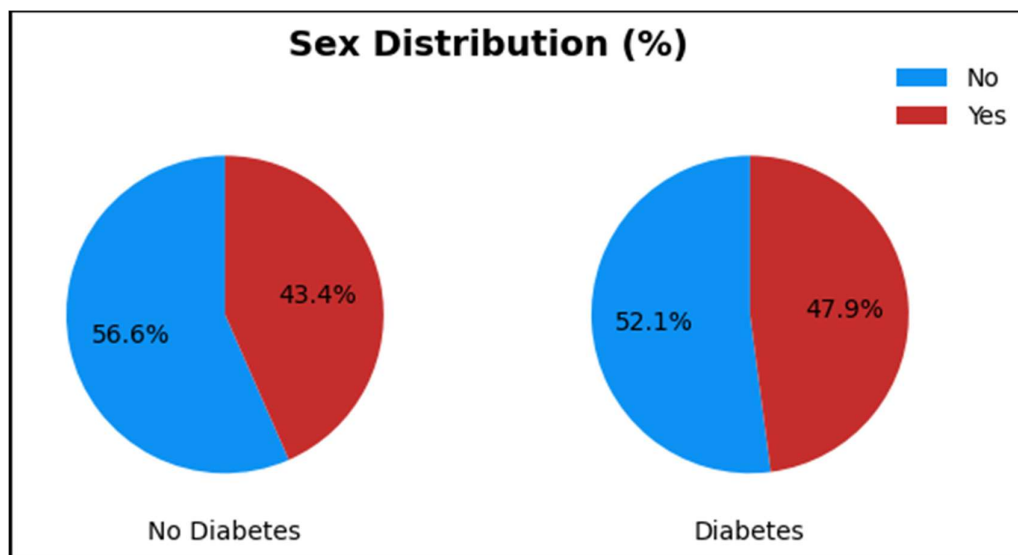


Figure 15 Pie Chart for Sex Distribution

2.3. Numerical Variable Trends

Histograms and mean value comparison across diabetes classes showed some distinct patterns:

Variable	Non-Diabetic Mean	Diabetic Mean
BMI	27.81	31.94
GenHlth	2.38	3.29
Age	7.81	9.38
PhysHlth	3.64	7.95

- Diabetic persons consistently showed a **higher BMI, older age, and lower general health.**
- Diabetic persons had more days of poor physical and mental health.
- The income and education level were marginally lower for diabetic cases.

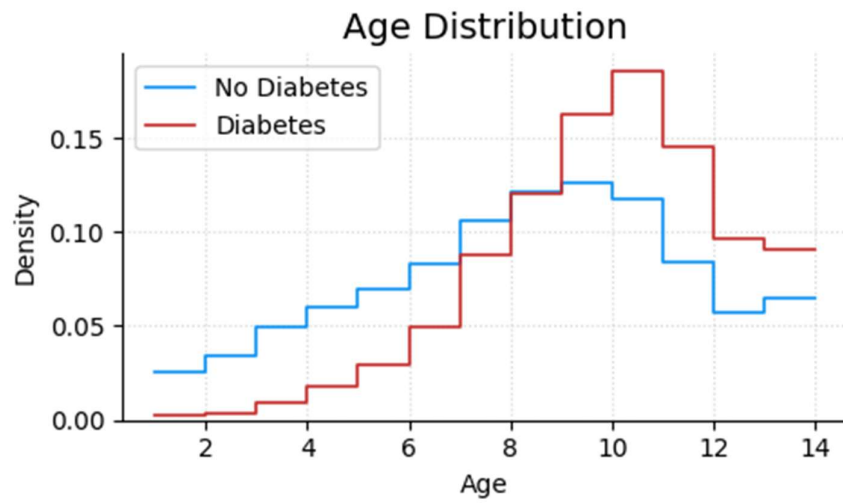


Figure 16 Histogram for Age Distribution

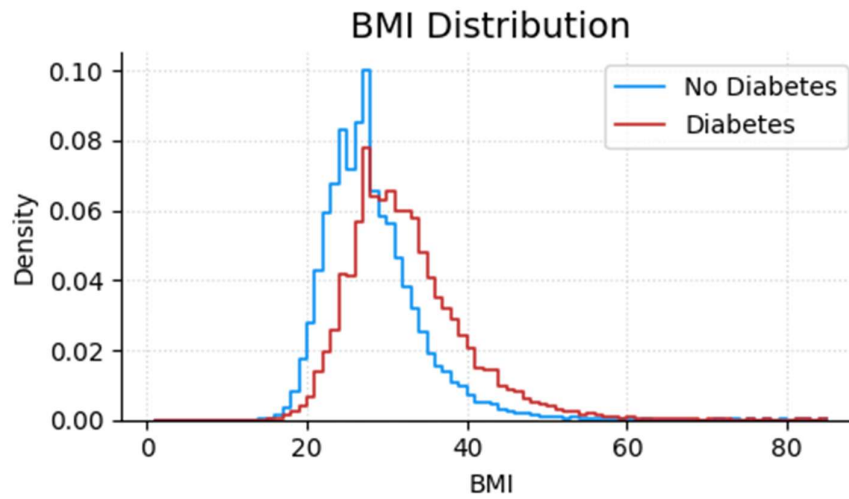


Figure 17 Histogram for BMI Distribution

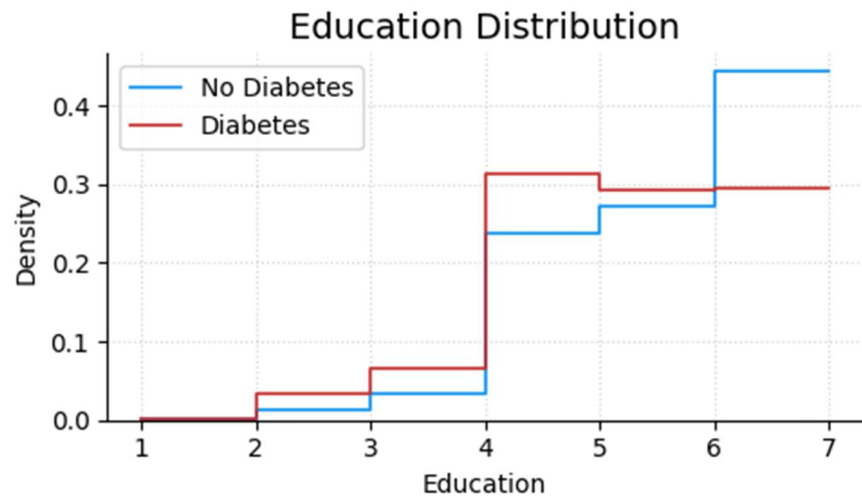


Figure 18 Histogram for Education Distribution

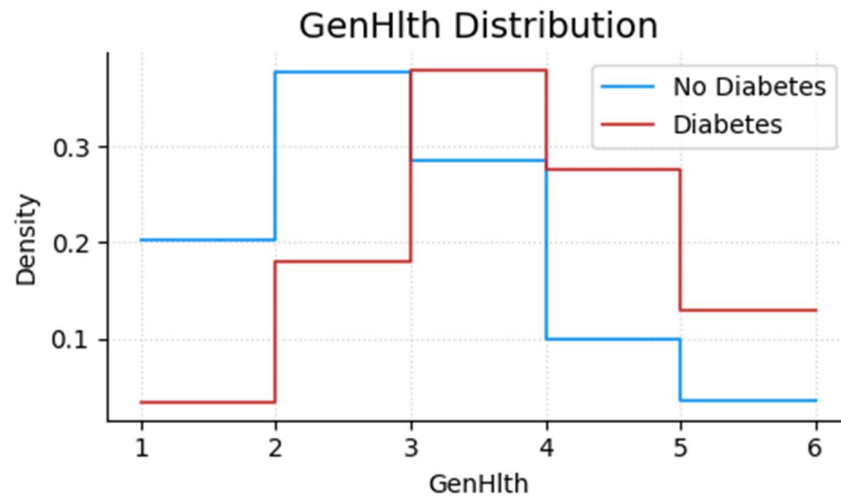


Figure 19 Histogram for GenHlth Distribution

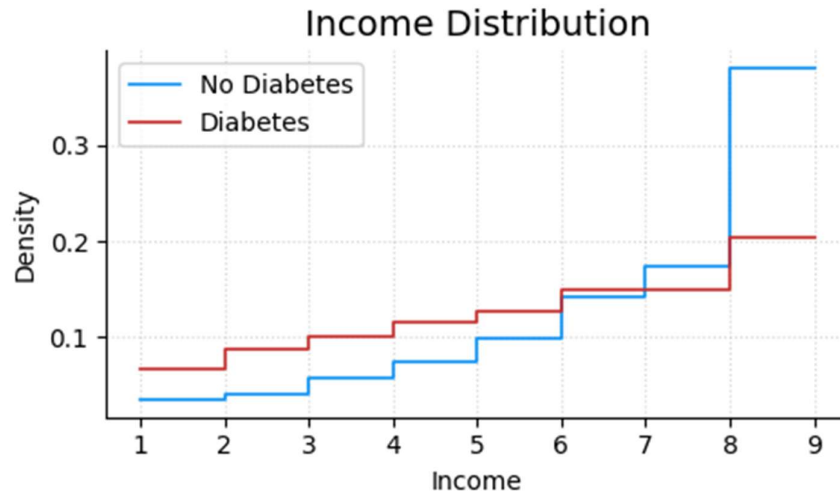


Figure 20 Histogram for Income Distribution

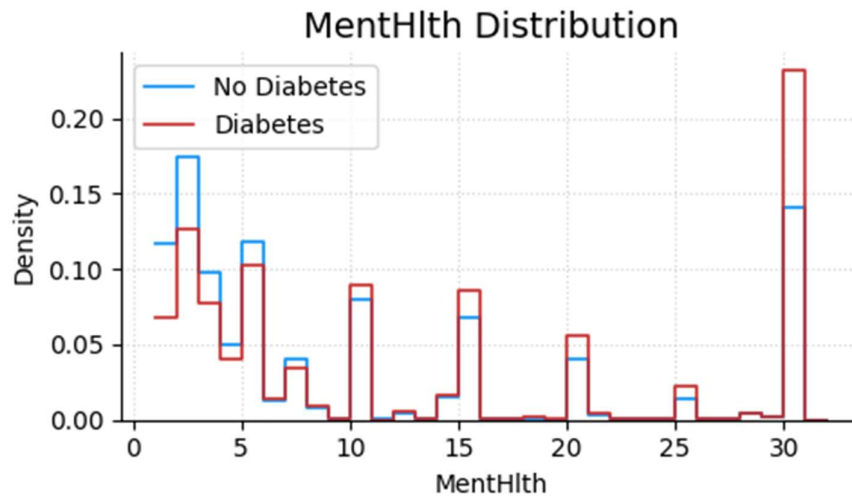


Figure 21 Histogram for MentHlth Distribution

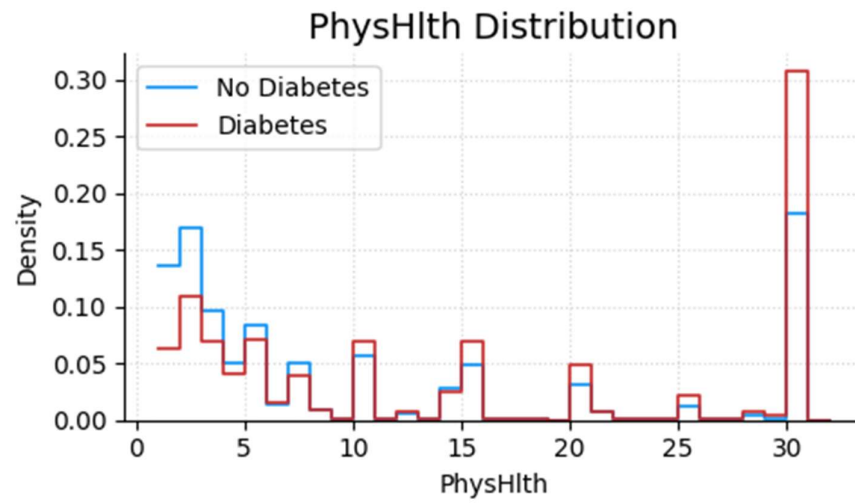


Figure 22 Histogram for PhysHlth Distribution

2.4. Correlation Analysis

A heatmap was generated using `df.corr()` to examine variable relationships. While overall correlations were modest, the strongest positive associations with `Diabetes_binary` include:

Feature	Correlation
BMI	0.28
GenHlth	0.25
Age	0.21
HighBP	0.15

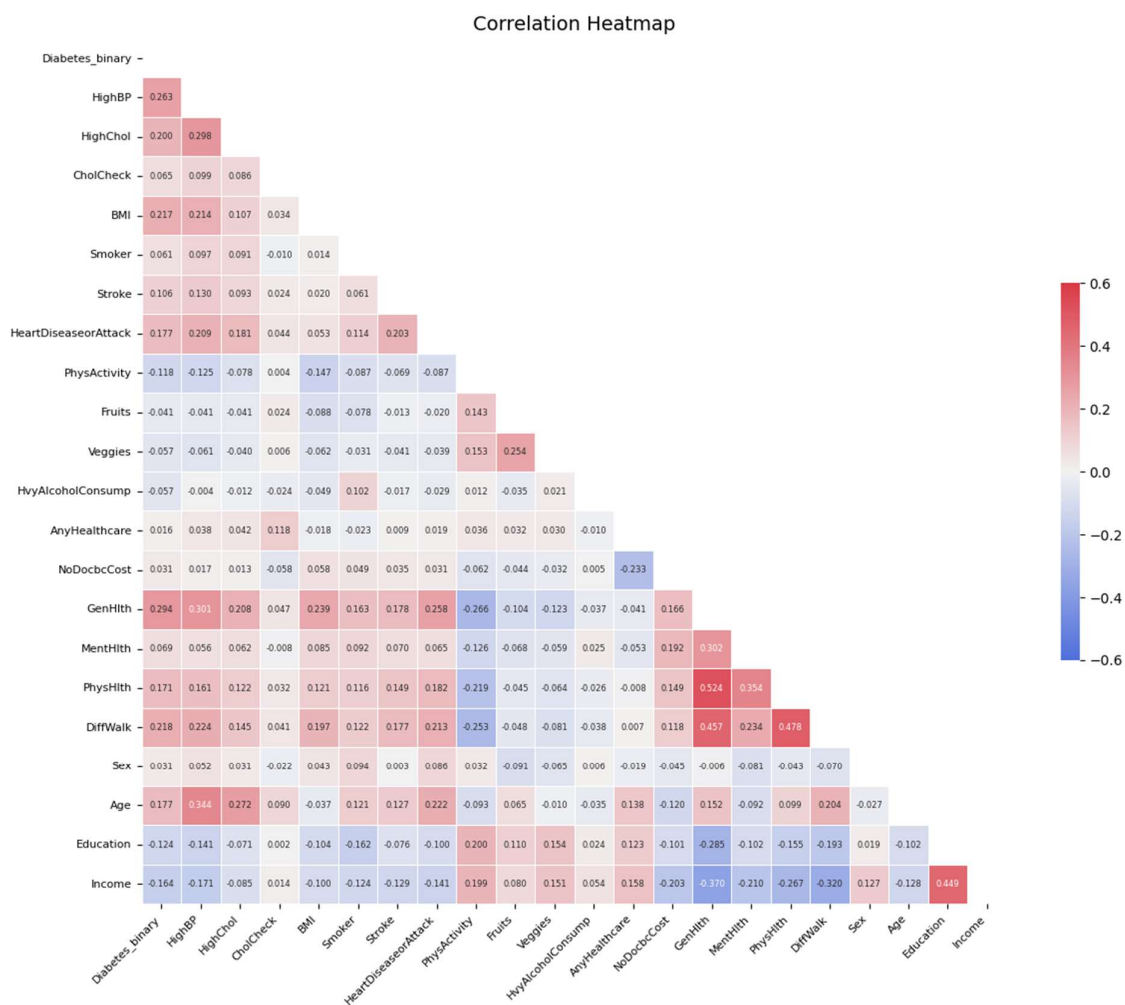


Figure 23 Correlation Heatmap

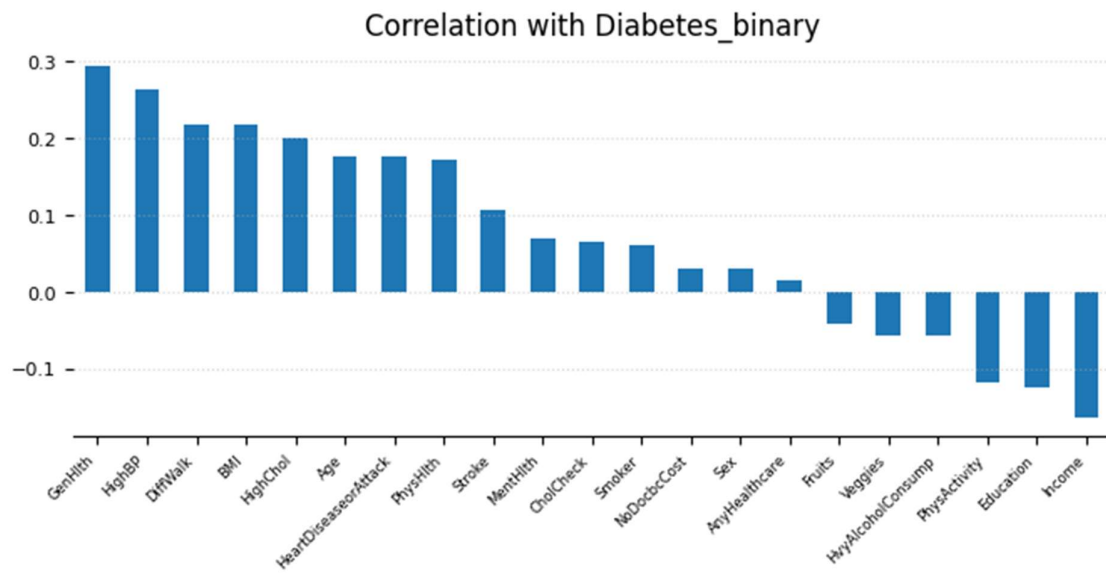


Figure 24 Bar Graph of the Correlation

2.5. Outlier Handling (BMI)

The boxplots revealed that BMI included extreme outliers exceeding clinical plausibility. The `find_outliers_iqr()` function was used to detect these values, and `winsorize_iqr()` clipped them to realistic boundaries based on IQR method:

- Lower Bound ≈ 12.0
- Upper Bound ≈ 44.0

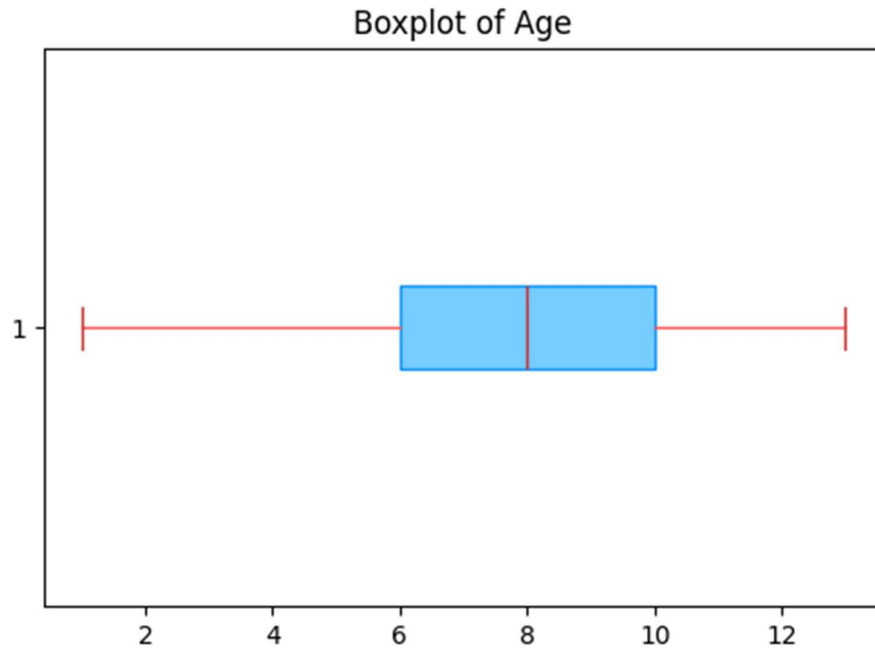


Figure 25 Boxplot for Age

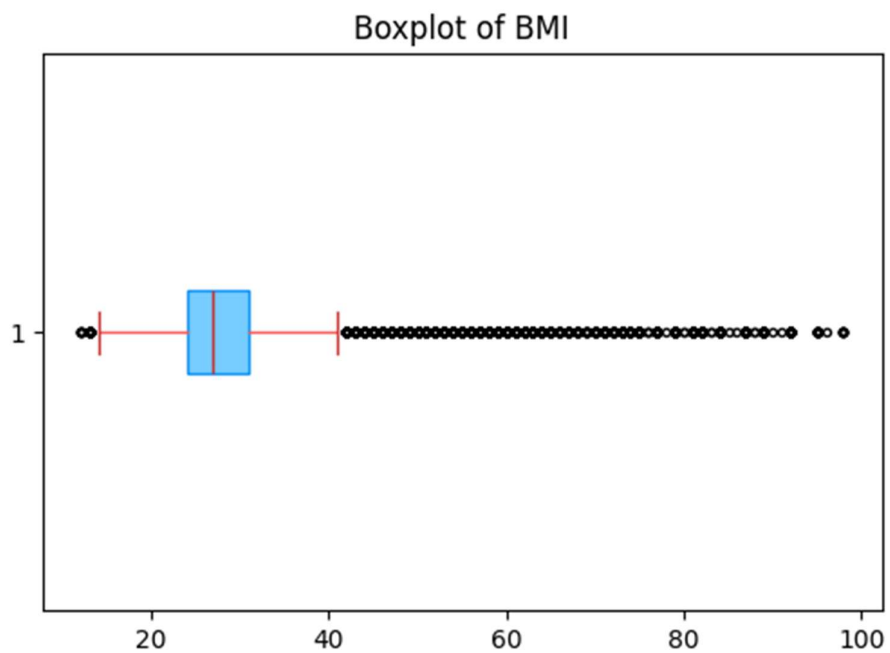


Figure 26 Boxplot for BMI

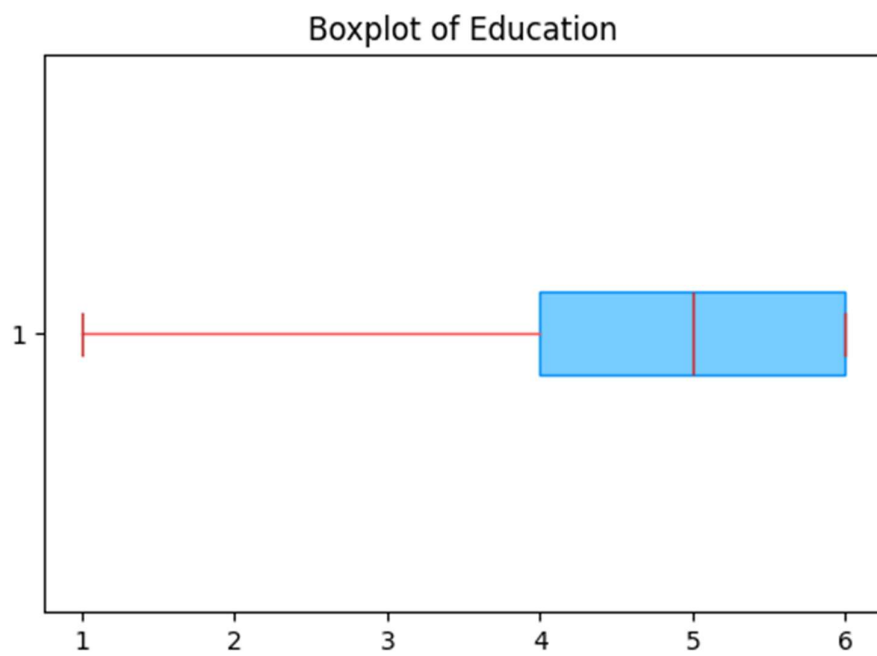


Figure 27 Boxplot for Education

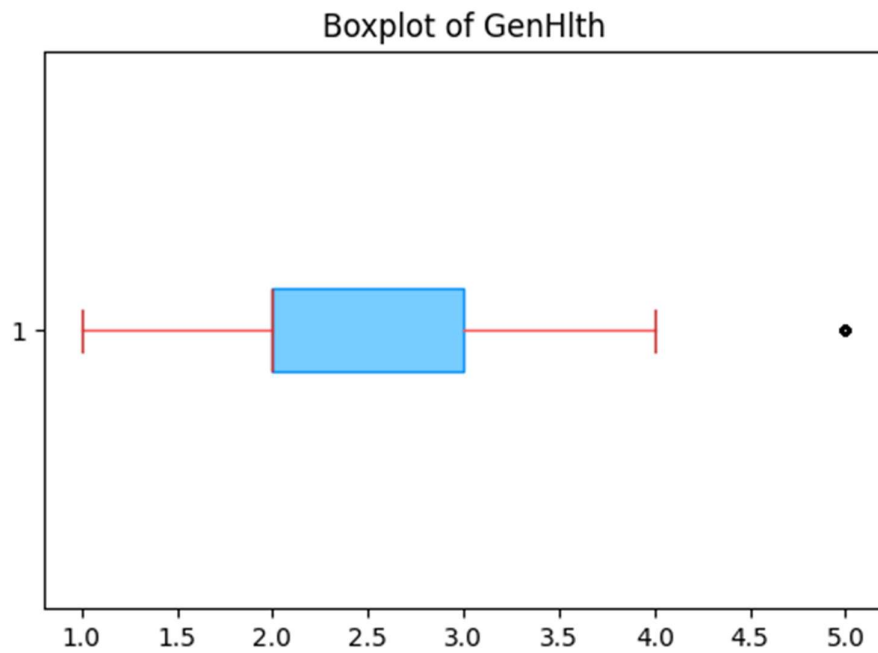


Figure 28 Boxplot for GenHlth

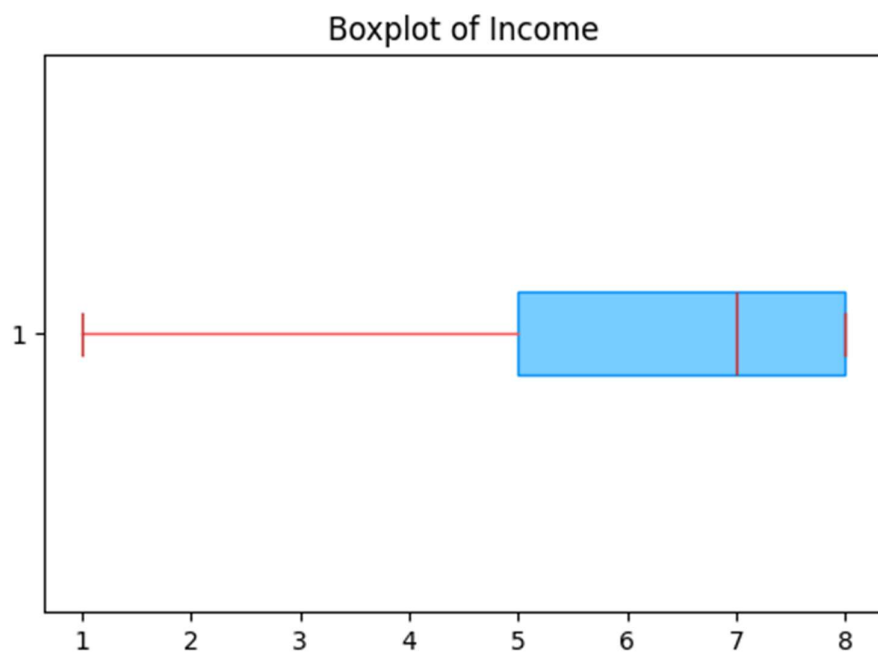


Figure 29 Boxplot for Income

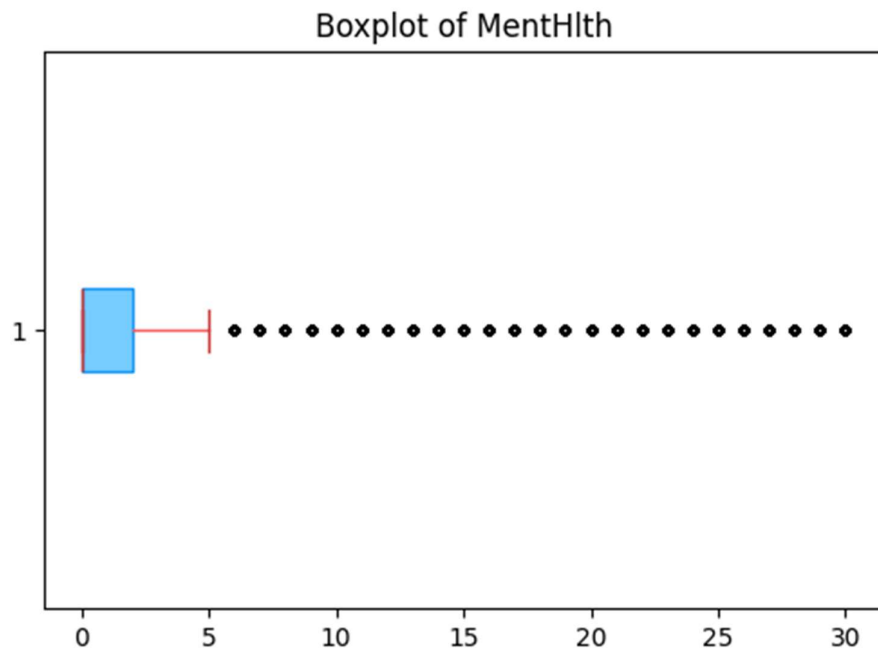


Figure 30 Boxplot for MentHlth

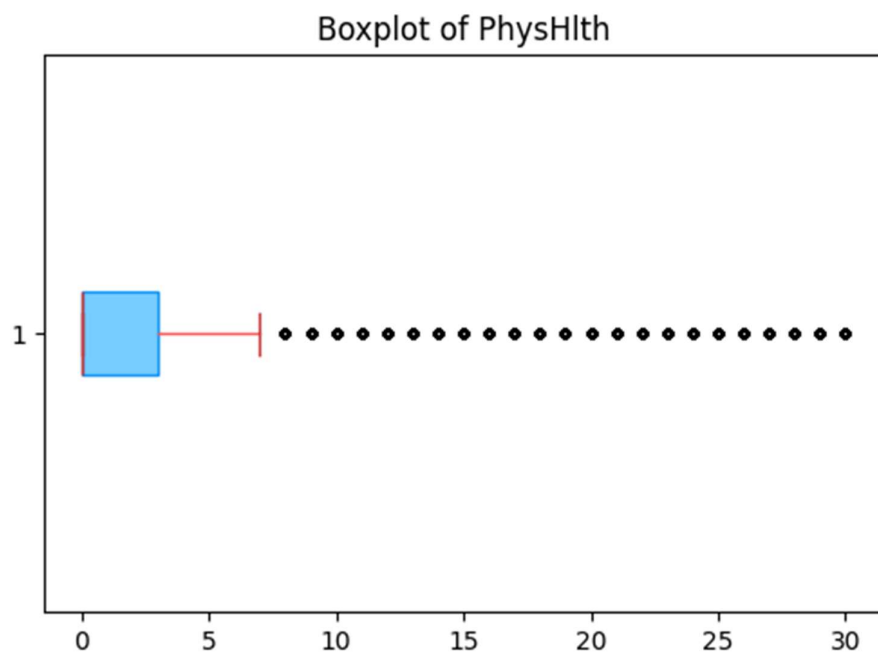


Figure 31 Boxplot for PhysHlth

2.6. EDA Conclusions

The dataset was comprehensive and clean, with little preprocessing or imputation needed. The exploratory analysis of the data indicated a diabetes occurrence of 13.93% as well as significant class imbalance, which was later addressed using SMOTE so the models would learn fairly. **BMI**, **GenHlth**, **Age**, and **HighBP** were statistically and clinically separated well between diabetic and non-diabetic patients, thus supporting their use as the best predictors for classification.

3. Feature Selection

Feature selection was performed to identify the most predictive variables for classifying the diabetes status. Several methods were used to ensure quality and agreement across statistical and model-based criteria.

3.1. Methods Used

Three techniques were applied:

- **SelectKBest (Univariate Selection)**
Based on `f_classif`, selecting the features with the strongest individual relationships to the target.
- **Recursive Feature Elimination (RFE)**
Using logistic regression to rank features by contribution to prediction.
- **Tree-Based Importance (RandomForestClassifier)**
Evaluating each feature's impact on decision-making across an ensemble of trees.

3.2. Selected Features

Across all three methods, four features consistently emerged as top predictors:

Feature	SelectKBest Score	RFE Rank	Tree Importance
BMI	12071.14	1	0.1762
GenHlth	19188.21	1	0.0702
Age	6579.66	1	0.1242
HighBP	15131.27	1	0.0436

3.3. Feature Selection Rationale

Numerous statistical and domain knowledge helped to select **BMI**, **GenHlth**, **Age**, and **HighBP** as key predictors. More importantly, these variables were the only ones given consistent priority during all approaches in **SelectKBest**, **Recursive Feature Elimination**, and **Tree-Based Importance**. These features are most clearly described in recent medical literature:

- **BMI (Body Mass Index)**

High BMI is the single most important modifiable risk factor in type 2 diabetes. Studies show that obesity is a major contributor to insulin resistance, with 90% of cases around the world being classified as type-2, and driven by lifestyle factors including poor diet and lack of activity (Liang, et al., 2025). Recent trends conclude that the increasing prevalence of diabetes globally is due to rising obesity levels (Collaborators, 2023).

- **GenHlth (Self-Rated General Health)**

Even though **GenHlth** is not a precise clinical measure, it indicates patient-perceived wellbeing and illustrates how it relates to comorbidities and chronic diseases. Individuals with diabetes in the examined dataset gave poorer health ratings than individuals without diabetes; an idea backed by public health models, which use survey data to develop models in which self-rated health was able to predict disease burden in the future, as well as healthcare utilization.

- **Age**

There is a major increase in diabetes risk in relation to age, especially regarding the age group 45+, as older age groups have the highest prevalence of "new onset" type 2 diabetes. This is due to cumulative exposure to metabolic risk factors and declines in insulin sensitivity (Liang, et al., 2025).

The mean age of diabetic subgroup was significantly older than non-diabetics, which validates age as a critical yet unmodifiable predictor.

- **HighBP (High Blood Pressure)**

Hypertension is one of the major comorbidities in diabetes at a rate of 66.3% across diagnoses and scales far greater than in non-glycemic participants (Wang, et al., 2025).

Poor blood pressure is common in diabetic populations - hypertension accounts for as much as 75% of cardiovascular risk in diabetes. Hypertension is such an important feature for risk modelling/prediction that it is promoted.

Together, these four categories look at a balance of lifestyle risk, clinical and self-reported features. Each of these features contributes compactly to the classification task and the development of models or reports to align with best evidence-based guidelines.

4. Modelling & Evaluation

In this section, we compared three supervised machine learning models—**Logistic Regression**, **Random Forest**, and **Gradient Boosting**—to predict diabetes risk using selected health indicators. Each model was trained using the same resampled dataset (SMOTE applied for class balance) and shared a common set of features: **BMI**, **GenHlth**, **HighBP**, and **Age**.

4.1. Comparison of Model Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
Logistic Regression	0.7208	0.2996	0.7506	0.4283	0.8094
Gradient Boosting	0.7538	0.7293	0.8072	0.7663	0.8322
Random Forest	0.7701	0.7632	0.7742	0.7687	0.8371

Observations:

- **Logistic Regression** achieved good recall (0.7506) but struggled in precision (0.2996), leading to a much lower F1-Score (0.4283). It tends to over-predict positives.
- **Gradient Boosting** delivered high recall (0.8072) and the best F1-Score (0.7663), making it excellent for prioritizing sensitivity.
- **Random Forest** slightly outperformed **Gradient Boosting** in overall accuracy (0.7701 vs. 0.7538) and ROC AUC (0.8371 vs. 0.8322), showing balanced performance across all metrics.

4.2. Comparison of Feature Importance

All models were trained on the same four features. The weight or importance of each feature was derived from each model's internal representation.

Feature	Random Forest	Gradient Boosting	Logistic Regression
GenHlth	0.3321	0.4042	0.6522
BMI	0.2889	0.3321	0.0983
HighBP	0.2193	0.1367	0.8711
Age	0.1651	0.1270	0.1853

Major Insights:

- **GenHlth** and **BMI** were consistently ranked highly in Random Forest and Gradient Boosting.
- **HighBP** was emphasized most by Logistic Regression, suggesting a stronger linear association in this model.
- **Age** had moderate influence in all models, validating its significance but also its non-dominance compared to lifestyle and comorbid factors.

4.3. Conclusion

Among the three models tested, **Random Forest** offered the best balance between **accuracy, precision, recall, and ROC AUC**, making it the most well-rounded performer. While **Gradient Boosting** showed slightly higher recall and F1-Score, it required more careful tuning and was computationally more intensive. **Logistic Regression**, although interpretable, suffered from low precision and is less suitable for imbalanced clinical data unless heavily tuned or regularized.

Therefore, **Random Forest was chosen as the final model** for deployment recommendation in this project, given its strong overall performance and robustness against overfitting.

5. Insights, Recommendations, and Ethics

5.1. Key Insights from the Analysis

Our comprehensive analysis of the Diabetes Health Indicators Dataset (Diabetes Health Indicator Dataset, 2022) revealed several key findings regarding diabetes risk prediction. The exploratory data analysis identified **BMI**, general health status (**GenHlth**), age, and high blood pressure (**HighBP**) as the most significant predictors of diabetes, with diabetics showing a considerably higher average **BMI** (31.94 vs 27.81) and lower self-reported health ratings. Across all models, feature importance analysis consistently ranked these four factors as the primary contributors, highlighting their clinical relevance. Among the algorithms evaluated, **Random Forest** performed the best overall, with an **AUC** of 0.8371 and an **accuracy** of 0.7701, while **Logistic Regression** was more interpretable but slightly less accurate. The class imbalance in our dataset (only 13.93% diabetic cases) necessitated **SMOTE** resampling to ensure fair model training. Notably, the high **recall** scores (0.7506-0.8072 across models) suggest that these methods are particularly effective at identifying true positive cases, which is vital for medical screening purposes. These findings demonstrate that clinical markers and lifestyle factors are crucial in diabetes risk assessment, while also emphasising the trade-offs between model complexity and interpretability in healthcare contexts.

5.2. Recommendations for Healthcare Applications

The outcomes of this study provide meaningful insights for healthcare providers and policymakers looking to lower diabetes risk through early intervention. Based on the findings, **Random Forest** is the best model for clinical deployment due to its balanced **accuracy** (0.7701) and **AUC** (0.8371), making it reliable for initial patient screening. Its capacity to detect high-risk patients, particularly those with an increased **BMI**, hypertension, or poor self-reported health, can aid in the prioritisation of preventive care resources. For example, clinics might incorporate this model into electronic health records (EHR) systems to identify at-risk patients during routine check-ups, requiring additional diagnostic tests or lifestyle counselling.

Meanwhile, **Logistic Regression**, even with slightly reduced predictive power, remains valuable for patient-facing applications where interpretability is essential. Its coefficients offer clear, explainable insights into risk factors (e.g., "High blood pressure increases diabetes risk by X percent"), which can motivate patients to adopt healthier behaviours. Community health programmes could utilise this model to initiate educational campaigns focused on modifiable factors such as physical inactivity or diet. A recent meta-analysis showed that a combination

of low-risk lifestyle behaviours, including maintaining a healthy body weight, following a nutritious diet, regular exercise, abstaining from or quitting smoking, and moderate alcohol intake, was linked to an 80% reduction in the risk of type 2 diabetes (Erika L Lundgrin, 2025).

5.3. Ethical Considerations

The use of predictive models for diabetes risk assessment involves various ethical issues that must be addressed in order to ensure responsible application in healthcare settings. First, algorithmic bias must be rigorously addressed because the model's performance may differ across demographic groups. For example, if the training data underrepresents specific races or socioeconomic backgrounds, it can result in inequalities in screening accuracy. Regular fairness audits should compare measures like recall and precision across subgroups to detect and mitigate such biases.

Finally, clinical over-reliance on AI is risky; even high-performing models should be used as decision-support tools, not as replacements for medical competence. Clear guidelines must ensure that physicians can override algorithmic recommendations when contextual factors (e.g., atypical symptoms) justify it. Proactive communication of the model's limitations, such as its reliance on self-reported data or failure to capture unmeasured genetic risks, can help minimise misuse while preserving patient autonomy. By addressing these ethical concerns, healthcare systems can use predictive analytics equitably and transparently.

Individual Component

1. KOO JUNN YUAN

1.1. Gradient Boosting Classifier

Gradient Boosting is an ensemble technique that involves creating models in a sequential manner where each learner tries to correct the errors made by the previous learner. It is a highly accurate and flexible technique that works particularly well when you have a non-linear relationship or a more complex dataset.

1.1.1. Model Configuration

The classifier was implemented using **Gradient Boosting Classifier** with the following parameters:

- `n_estimators = 100`
- `learning_rate = 0.1`
- `max_depth = 3`
- Resampled data using **SMOTE** for class balance
- Selected features: **BMI, GenHlth, HighBP, and Age**

1.1.2. Evaluation Metrics

Metric	Default Threshold	Threshold = 0.6
Accuracy	0.7538	0.5849
Precision	0.7293	0.7683
Recall	0.8072	0.7008
F1-Score	0.7663	0.7330
ROC AUC	0.8322	-

At the default threshold, the model achieved strong performance across all key metrics, particularly in **Recall (0.8072)** and **AUC (0.8322)**, indicating high sensitivity and ranking ability.

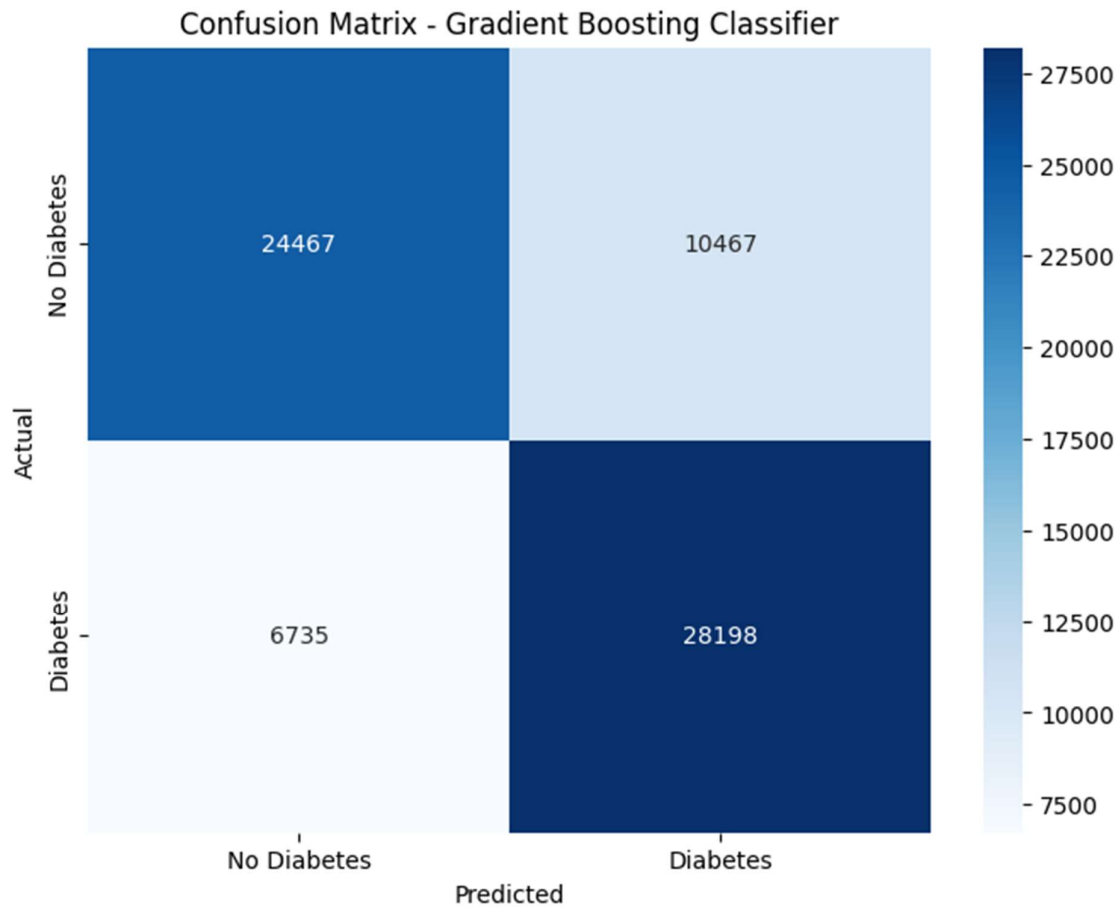


Figure 32 Confusion Matrix for Gradient Boosting Classifier

1.1.3. Feature Importance

Feature	Importance
GenHlth	0.4042
BMI	0.3321
HighBP	0.1367
Age	0.1270

Feature importance aligns with clinical findings. **GenHlth** and **BMI** demonstrated the highest predictive power, consistent with domain knowledge and supporting literature.

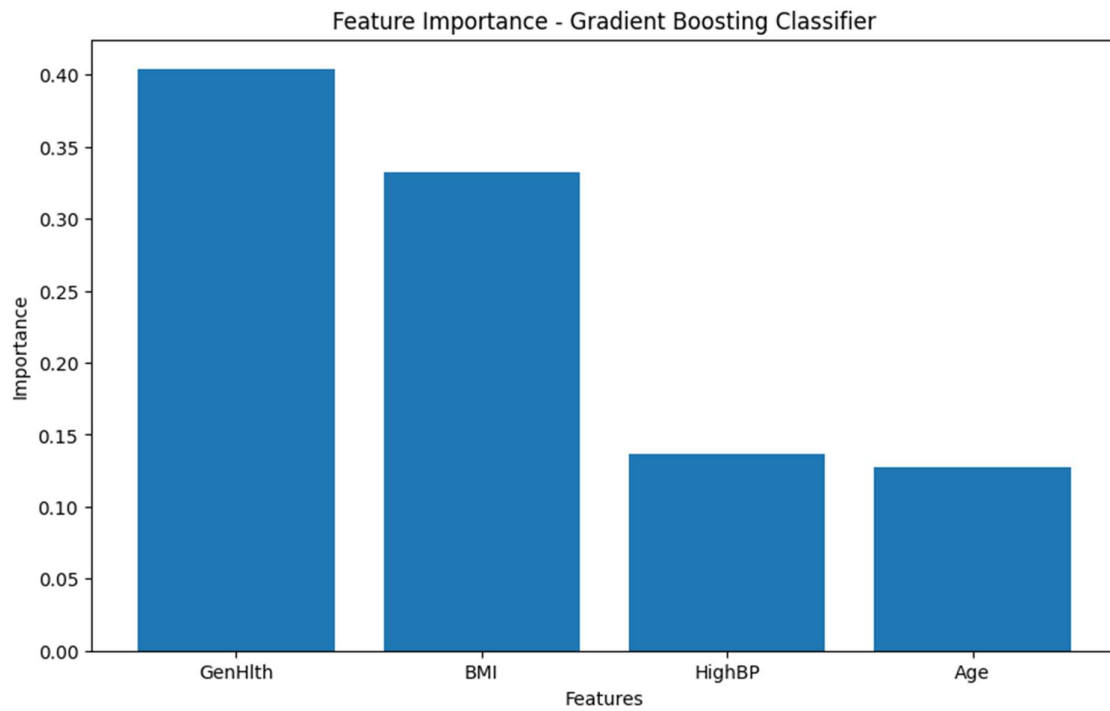


Figure 33 Graph of Feature Importance for Gradient Boosting Classifier

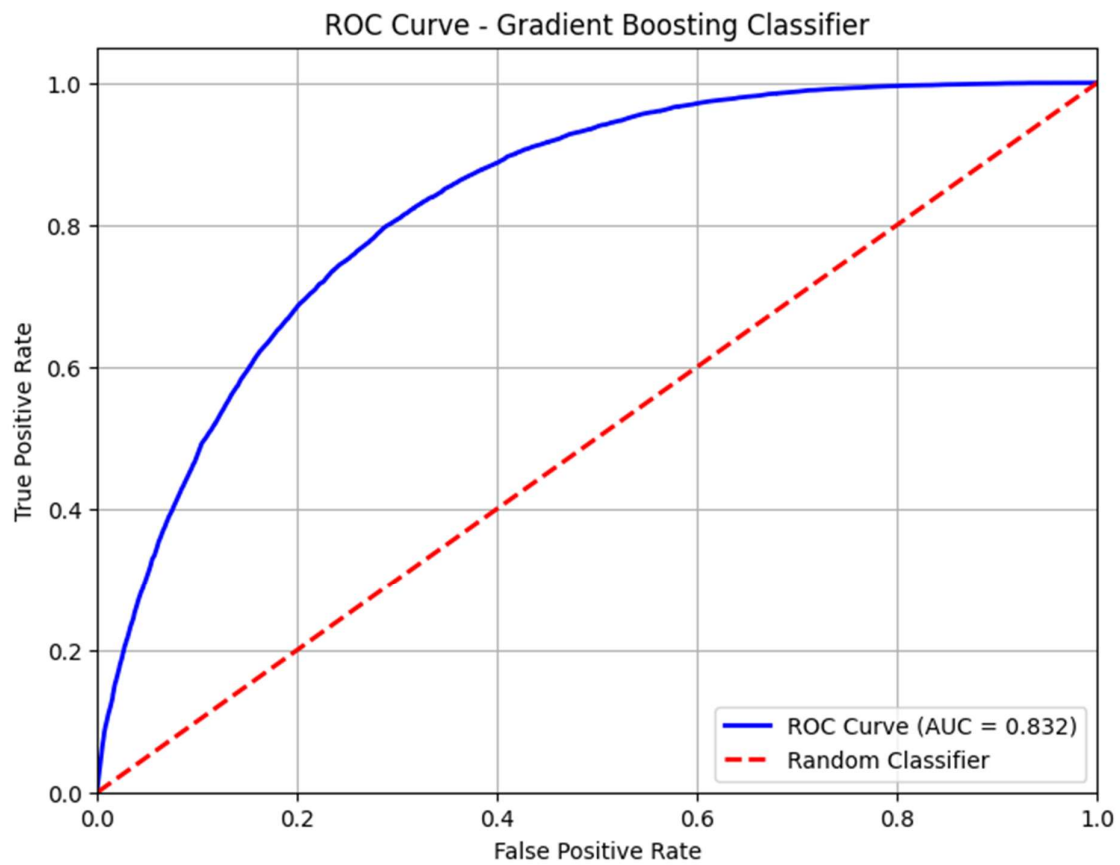


Figure 34 ROC Curve for Gradient Boosting Classifier

1.1.4. Comparative Analysis

Strengths of Gradient Boosting:

- **High recall** and **F1-score** suitable for detecting diabetes risk.
- Strong AUC performance suggests effective probability calibration.

Limitations:

- More sensitive to hyperparameter choices.
- Higher computational cost than **Logistic Regression**.
- Slight reduction in precision at lower thresholds, requiring careful tuning.

1.1.5. Reflection

Gradient Boosting was a very strong option with excellent probability ranking and sensitivity in diabetes classification. It was important to consider that in a clinical setting, it may be costly to miss a positive case, so **high Recall** would be more appealing. However, since **Random Forest** had demonstrated consistent **accuracy** and was easily deployed, the team ultimately agreed that **Random Forest** was the best model to proceed with.

2. KUBERAN A/L KALAI SELVAN

2.1. Random Forest Classifier

Random Forest is an ensemble machine learning algorithm that constructs multiple decision trees during training time and outputs the mode of the classes for classification tasks. It is robust against overfitting, handles non-linear features effectively, and provides reliable feature importance rankings. Its versatility makes it a strong candidate in medical data classification problems, especially where interpretability and performance are both important.

2.1.1. Model Configuration

The classifier was implemented using **Random Forest Classifier** with the following parameters:

- `n_estimators = 100`
- `max_depth = 5`
- `random_state = 42`
- `criterion = 'gini'`
- Data was **resampled using SMOTE** for class balance
- Selected features: **BMI, GenHlth, HighBP, Age**

2.1.2. Evaluation Metrics

Metric	Default Threshold (0.5)
Accuracy	0.7701
Precision	0.7632
Recall	0.7742
F1-Score	0.7687
ROC AUC	0.8371

At the default threshold of 0.5, **Random Forest** demonstrated **balanced precision and recall**, indicating reliable detection of diabetic individuals while minimizing false positives. The **ROC AUC of 0.8371** highlights excellent model discrimination capabilities. Unlike the other models, I used the default threshold because the model was good enough after doing some major testing

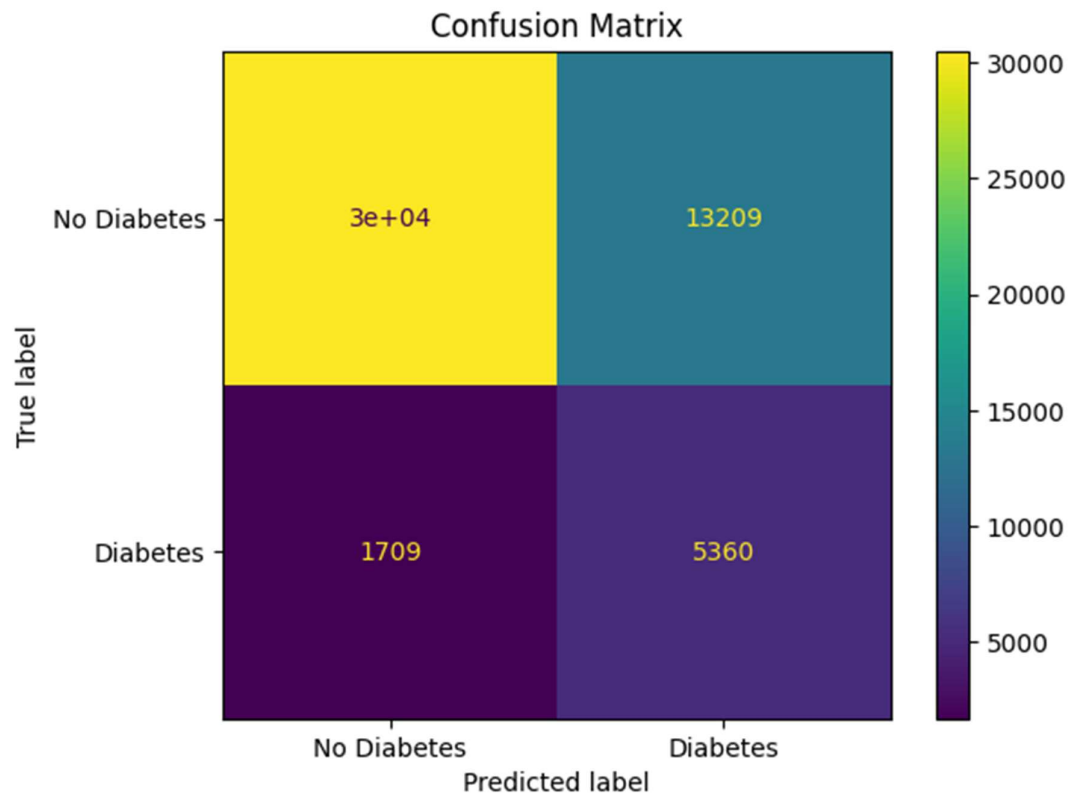


Figure 35 Confusion Matrix for Random Forest

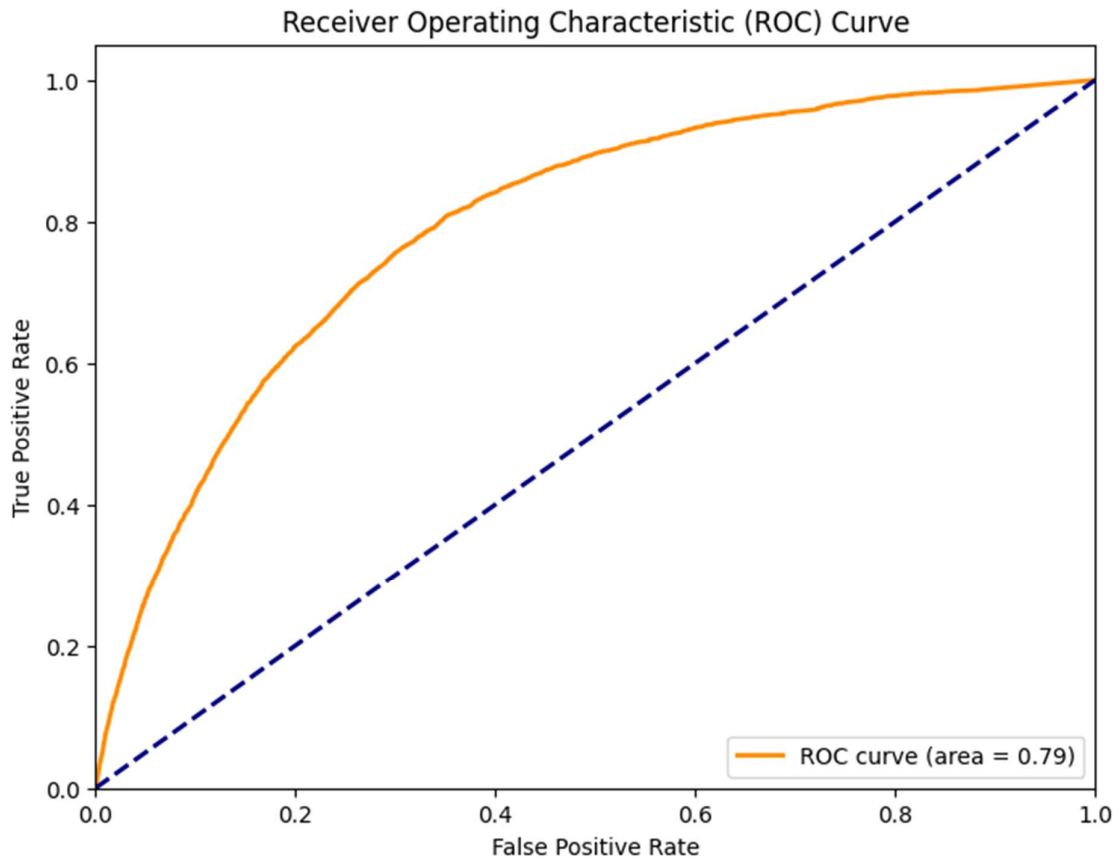


Figure 36 ROC Curve for Random Forest

2.1.3. Feature Importance

Feature	Importance
GenHlth	0.3321
BMI	0.2889
HighBP	0.2193
Age	0.1651

Feature importance aligns closely with domain knowledge and EDA findings. **GenHlth** and **BMI** were the most influential predictors, further confirming their clinical relevance in predicting diabetes risk.

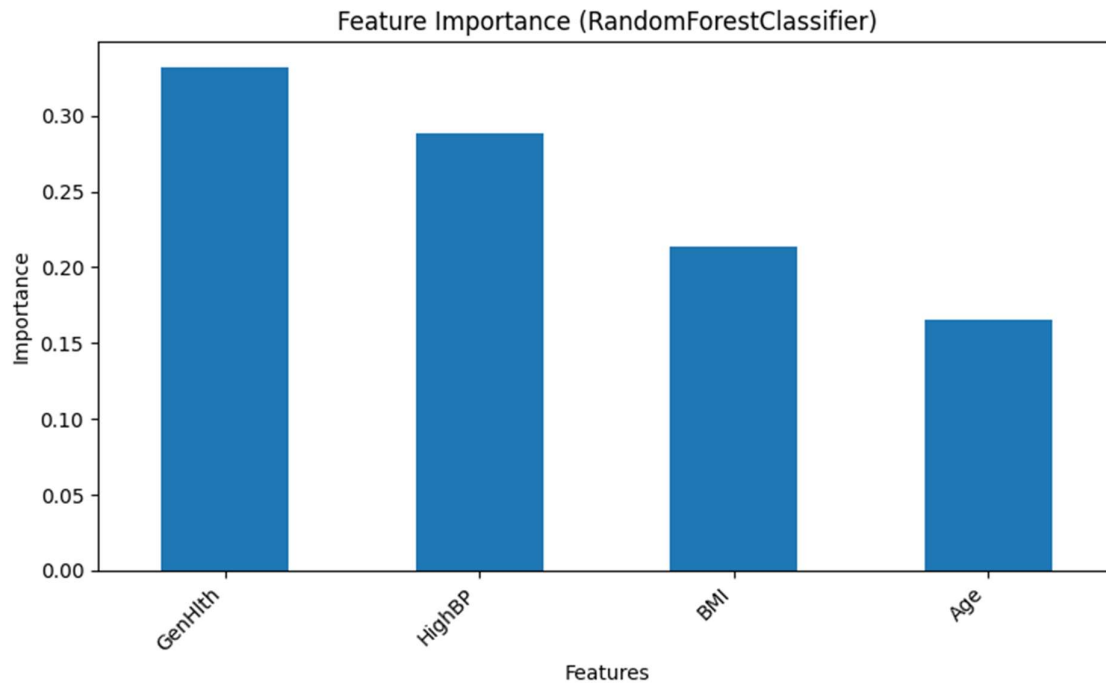


Figure 37 Graph of Feature Importance for Random Forest

2.1.4. Comparative Analysis

Strengths of Random Forest:

- Provides intuitive feature importance metrics.
- Delivers **strong accuracy and ROC AUC**, making it suitable for clinical deployment.

Limitations:

- Less interpretable compared to simpler models (like logistic regression).
- May not perform well if too many irrelevant features are included (but this was addressed with proper feature selection).

2.1.5. *Reflection*

I found **Random Forest** to be a powerful and dependable algorithm for medical prediction tasks. It provided not only solid predictive performance but also interpretability through feature importance. In the context of diabetes risk, this is crucial for gaining trust from healthcare professionals. The model balanced sensitivity and specificity well, which is vital when predicting chronic diseases where early intervention can save lives. Although **Gradient Boosting** slightly edged out in AUC, **Random Forest** was more **computationally efficient** and **less prone to hyperparameter sensitivity**, making it a practical choice for real-world deployment.

3. LOW WEI HAO

3.1. Logistic Regression

Logistic regression is a supervised machine learning algorithm used to solve classification problems. In contrast to linear regression, which predicts continuous values, this method predicts the likelihood that an input belongs to a specific class. It is used for binary classification, which has two possible outputs: Yes/No, True/False, or 0/1. It uses the sigmoid function to convert inputs into a probability value between 0 and 1.

3.1.1. Model Configuration

The classifier was implemented using **Logistic Regression** with the following parameters:

- `test_size = 0.2`
- `random_state = 42`
- `stratify = y`
- Data was resampled using SMOTE for class balance
- Selected features: **BMI, GenHlth, HighBP, and Age**

3.1.2. Evaluation Metrics

Metric	Default Threshold	Threshold = 0.5
Accuracy	0.7208	0.7208
Precision	0.2996	0.2996
Recall	0.7506	0.7506
F1 - Score	0.4283	0.4283
ROC - AUC	0.8094	-

At the default threshold, the model achieved strong performance across all key metrics, particularly in **Recall (0.7506)** and **AUC (0.8094)**, indicating high sensitivity and ranking ability.

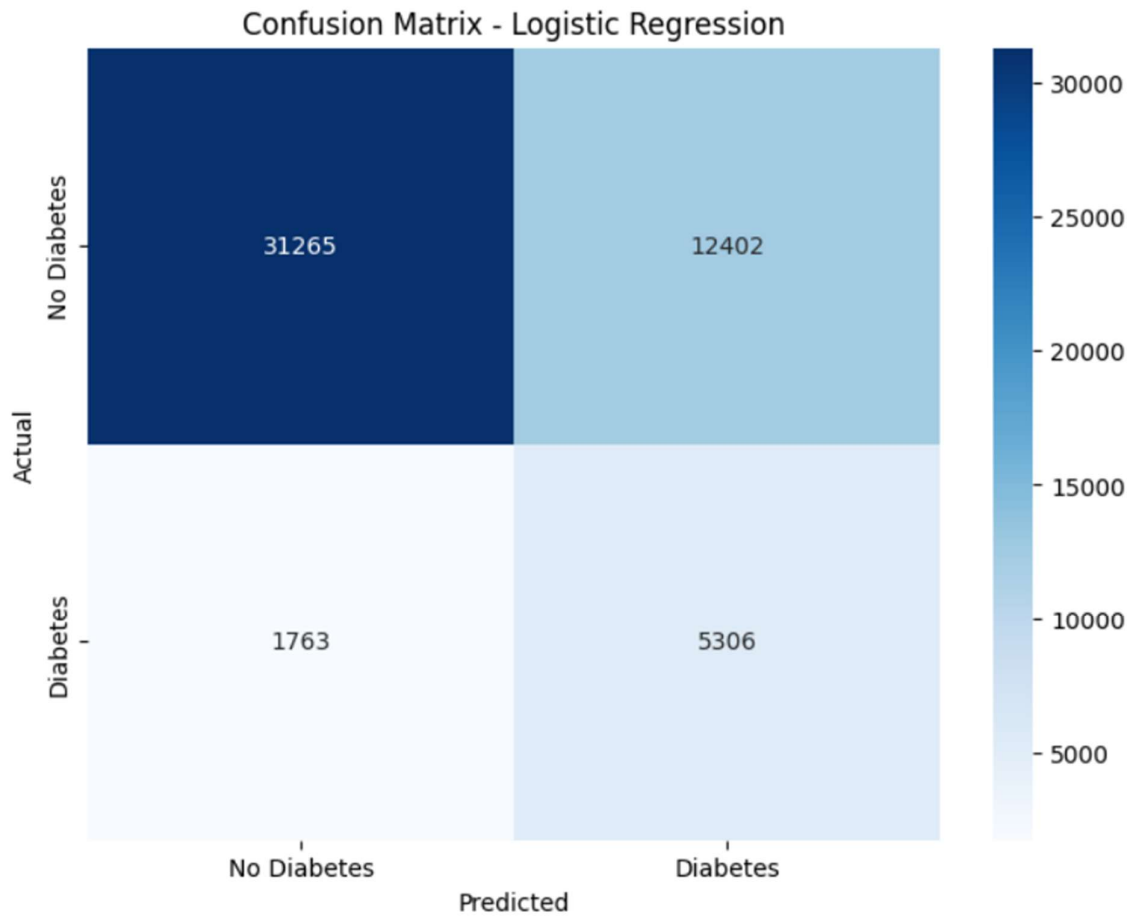


Figure 38 Confusion Matrix for Logistic Regression

3.1.3. Features Importance

Feature	Importance
HighBP	0.8711
GenHlth	0.6522
Age	0.1853
BMI	0.0983

Feature importance aligns with clinical findings. **HighBP** and **GenHlth** demonstrated the highest predictive power, consistent with domain knowledge and supporting literature.

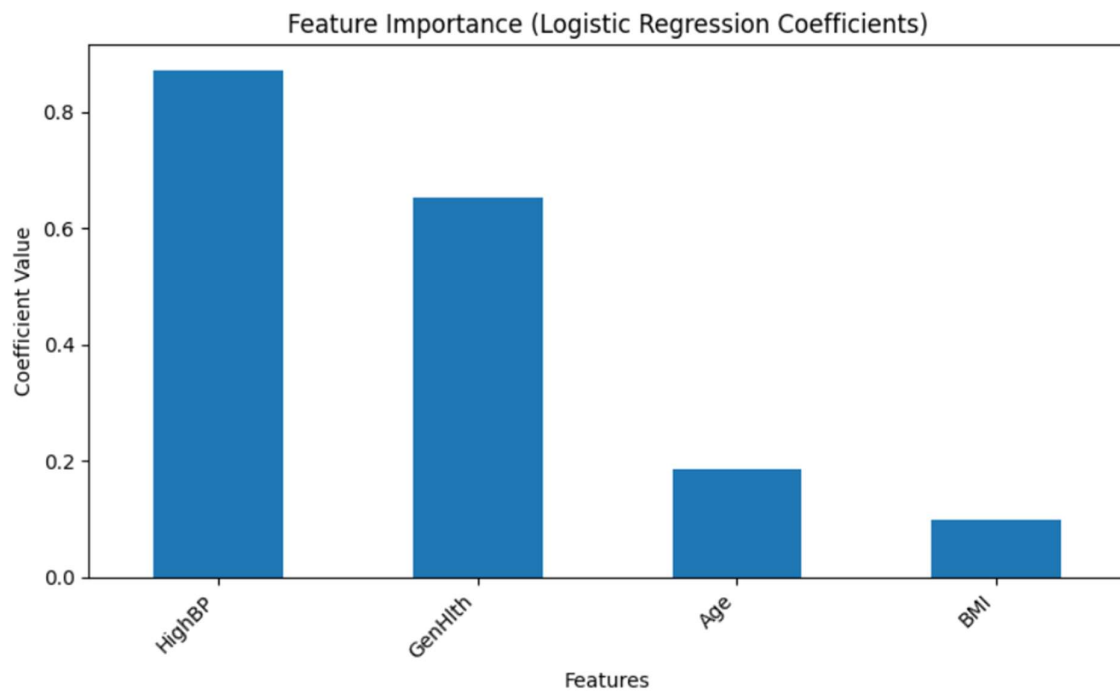


Figure 39 Feature Importance for Logistic Regression

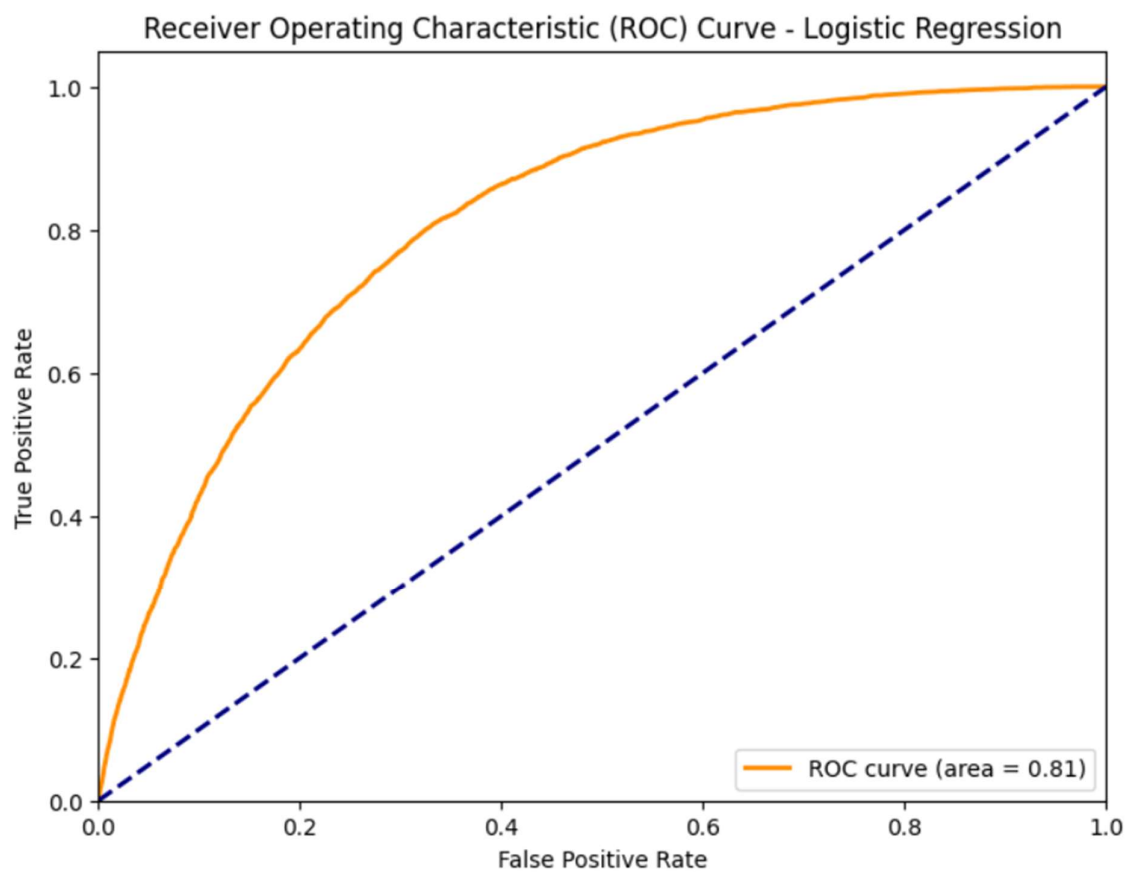


Figure 40 ROC Curve for Logistic Regression

3.1.4. *Comparative analysis*

Strengths of Random Forest:

- Provides clear coefficients for each feature, showing the direction and strength of their relationship with diabetes risk.
- Directly estimates the probability of diabetes, allowing flexible threshold adjustments for clinical use.

Limitations:

- May underperform if relationships between features and diabetes risk are highly nonlinear.
- Produces more false positives compared to Random Forest (as seen in the lower precision score).

3.1.5. *Reflection:*

Logistic Regression is an effective baseline model because of its simplicity and interpretability, which are important in healthcare settings where explaining predictions is critical. However, given its lesser precision, ensemble approaches such as **Random Forest** or **Gradient Boosting** may be more reliable for clinical deployment, particularly where **accuracy** and **AUC** are prioritised. Future enhancements may include feature engineering or hybrid modelling to boost speed while maintaining interpretability.

6. Personal Reflection

1. KOO JUNN YUAN

This project has given me clarity on effective teamwork and leadership, particularly in leading a team through a complex technical project. Not only did I grow personally through the module, but it has also introduced me to fundamental data science techniques - data cleaning, exploratory analysis, model imbalance, and performance measures for predictive models.

As I built and tested machine learning models, I was able to learn through practice about developing a solution within a healthcare context. The ability to consider the healthcare context was particularly valuable as it allowed me to think about the operational challenges and social impact of identifying and detecting early disease.

Overall, this assignment has advanced my analytical ability and improved my damage control and collaboration skillset, which I hope to apply in my future work in data informatics and in solving problems in the real world.

2. KUBERAN A/L KALAI SELVAN

This project was a very valuable experience to learn the data analytics techniques in the context of a real-world health issue of predicting risk for diabetes. I learned a great deal from the process of completing the detailed EDA, feature selection, and model tuning. I already had some awareness of how decision-making through data (data-driven decisions) could positively impact public health. I liked the Random Forest model for its quality as a model that effectively combined accuracy and interpretability. I enjoyed working with my team to further our understanding of ethical issues in predictive modelling and the issue of bias and responsible use of models. In general, I learned a great deal about analytics, as it relates to the bigger picture of clinical decision-making, which included refreshing my technical skills while reinforcing the importance of working in collaboration with others

3. LOW WEI HAO

This study gave excellent hands-on experience in using machine learning to forecast diabetes risk, a significant real-world health concern. I improved my technical abilities and understanding of healthcare analytics by working through the entire EDA, feature engineering,

and model optimisation process from start to finish. I was particularly impressed with how the Random Forest model matched accuracy and interpretability, providing clear insights into critical risk factors such as BMI and age. Collaborating with teammates highlighted the need of matching technical approaches with clinical competence while addressing ethical concerns such as bias reduction. Beyond honing my Python and scikit-learn skills, this work demonstrated how data science can produce meaningful health benefits by combining rigorous analysis, cross-disciplinary cooperation, and responsible implementation.

7. Appendix

7.1. Appendix A: Dataset Inspection & Preparation

```
df.info()
```

```
df.describe().T
```

```
df.isna().sum()
```

Confirms dataset completeness, no missing values, and numeric data types.

```
df['Diabetes_binary'] = df['Diabetes_binary'].replace({2: 1})
```

Reclassifies pre-diabetic cases to reflect clinical grouping under the diabetic category.

7.2. Appendix B: Outlier Detection & Winsorization

```
def winsorize_iqr(df, col, k=1.5):  
    Q1 = df[col].quantile(0.25)  
    Q3 = df[col].quantile(0.75)  
    IQR = Q3 - Q1  
    lower_bound = Q1 - k * IQR  
    upper_bound = Q3 + k * IQR  
    return df[col].clip(lower=lower_bound, upper=upper_bound)
```

```
df['BMI'] = winsorize_iqr(df, 'BMI')
```

Handles extreme BMI values to avoid model distortion.

7.3. Appendix C: Feature Selection

```
from sklearn.feature_selection import SelectKBest, f_classif

k_best = SelectKBest(score_func=f_classif, k=10)
X_train_kbest = k_best.fit_transform(X_train, y_train)
selected_features_kbest =
X_train.columns[k_best.get_support()]

from sklearn.feature_selection import RFE
from sklearn.linear_model import LogisticRegression
```

Applies univariate statistical test to select the top predictors.

```
rfe = RFE(estimator=LogisticRegression(solver='liblinear'),
n_features_to_select=10)
X_train_rfe = rfe.fit_transform(X_train, y_train)
selected_features_rfe = X_train.columns[rfe.support_]
```

Ranks features by contribution via Recursive Feature Elimination.

7.4. Appendix D: Sampling Strategy

```
from imblearn.over_sampling import SMOTE

smote = SMOTE(random_state=42)

X_train_resampled, y_train_resampled =
smote.fit_resample(X_train_selected, y_train)
```

Addresses class imbalance using synthetic oversampling.

7.5. Appendix F: Evaluation Metrics

```
from sklearn.metrics import f1_score, roc_auc_score
```

```
f1 = f1_score(y_test, y_pred)
```

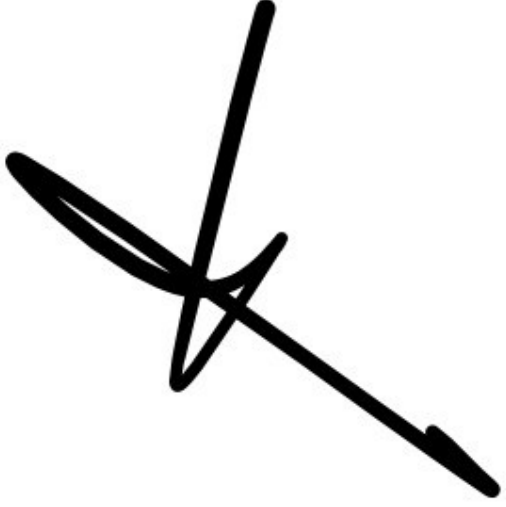
```
roc_auc = roc_auc_score(y_test, y_pred_proba)
```

Computes performance metrics for medical classification.

Google Colab Notebook link:

https://colab.research.google.com/drive/1Q5Jz7B5ql3e9fE5agYmD5H_Kn2nt-zUd?usp=sharing

8. Workload Matrix

Members	Tasks	Signature
KOO JUNN YUAN	• Tested and trained the Gradient Boosting Classifier model. • Completed the individual component • Performed the dataset inspection and preparation • Performed Exploratory Data Analysis (EDA) • Designed the overall format of the report	
KUBERAN A/L KALAI SELVAN	• Tested and trained the Random Forest Classifier model. • Completed the individual component • Conducted the feature selection and data sampling • Completed the model comparison • Made the Streamlit UI	
LOW WEI HAO	• Tested and trained the Logistic Regression model. • Completed the individual component • Conducted research • Complete the insights, recommendations, and ethical considerations	

9. References

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